A Complex Namespace for the rho Calculus

Paired messages, two quotations, and an intertwining COMM

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Version 1 — a record of the core calculus

Abstract

This note records, in one place, the core of a variant of the rho calculus designed so that the namespace carries the structure of a complex scalar field rather than that of a free commutative monoid. Three changes are made to the term language and one to the reduction relation. Messages carry *pairs* of processes, so that a message is a formal difference and negation is structural. Names are quotations of pairs, so that negation is an operation on *addresses* and not merely a decoration on sends. There are two quotation operators and, of necessity, two dropping operators; the two quotations are read as a real and an imaginary sector. Communication *intertwines* the sectors, so that a communication event is multiplication by the imaginary unit, and communication between participants of opposite polarity exchanges the components of the received pair, so that opposite polarity contributes the sign. The note states what has been settled, states the design forks that have not, and enumerates the obligations the design incurs. It proves almost nothing. Its purpose is to fix notation and to make the open questions precise enough to be worked on.

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1 What this note is

1.1 Scope

The present document is a *record*, not an argument. It writes down the term language, the structural theory and the reduction relation of a variant calculus that has emerged from a sequence of design decisions, together with the reasons each piece is present. It does not develop the semantics that motivated the design; that semantics — an interpretation of the operations of quantum mechanics over a reflective process calculus, with processes as vectors, names as scalars, evaluated contexts as covectors and contexts as operators — is the subject of separate work, and is recalled here only far enough to explain why each feature was added.

Two things are deliberately excluded. The note gives no metatheory: no subject reduction, no congruence result, no decidability. And it gives no semantics: no interpretation map, no inner product, no dagger beyond the one line needed to say which operation the dagger is built from. Both are named as obligations in Section 9.

1.2 Status markers

Throughout, claims carry one of three markers.

SETTLED The design decision has been taken and the consequences worked through. It may still be wrong, but it is not open.

FORK Two or more readings are consistent with what has been settled, and the choice has not been made. The alternatives are stated.

OPEN An obligation the design incurs and does not discharge.

1.3 Notational rules

The calculus is called the `rho` calculus and is never written with the Greek letter that follows pi; the name abbreviates *reflective higher order*, and the transliteration is the pun. Quotation is written @ and dropping $*$; corner quotes are not used.

The design discussions that produced this calculus wrote the pair constructor as an infix asterisk, $Q_p * Q_n$. That notation collides with the dropping operator, so pairs are written here with angle brackets, $\langle Q_p, Q_n \rangle$, and the reader translating from those discussions should read $\langle Q_p, Q_n \rangle$ wherever $Q_p * Q_n$ appears there.

2 Why the namespace is expanded

The calculus below is the minimal response to four defects, each of which is a defect of the *unexpanded* namespace. They are recalled because each feature of the design is answerable to one of them, and a reader who forgets the defects will find the features arbitrary.

The scalars have no additive inverses. In the ordinary rho calculus, structural congruence is associativity, commutativity and a unit for parallel composition. Consequently $\mathsf{Proc}/\equiv$ is the free commutative monoid on the non-parallel components of processes, and Name, being $\mathsf{Proc}/\equiv$ transported through quotation, is that same free commutative monoid. A free commutative monoid has no inverses and no cancellation. Whatever else it can express, it cannot express the vanishing of a sum of nonzero contributions. Since destructive interference is exactly such a vanishing, no coefficient structure carried by names can produce it.

The multiplicative and additive units collide. Name multiplication is $@P \cdot @Q := @(P \mid Q)$, whose unit is $@\mathsf{0}$. Any addition on names built from summation also has $@\mathsf{0}$ for its unit. The rig law $0 \cdot x = 0$ then fails, since $@\mathsf{0} \cdot x = x$. There is no absorbing element, and hence no rig.

There is no conjugation. Parallel composition is commutative, so any pairing built from it is symmetric rather than Hermitian, and there is no involution available on the coefficients to repair this. An involution must come from somewhere; the unexpanded calculus has none of order two acting on scalars.

Multiplication by a monomial is a shift, not a scaling. This is the sharpest of the four. In the free commutative monoid, multiplying a name by another name is injective and never decreases anything; it permutes basis elements rather than combining them. Any attempt to realise the imaginary unit as an ordinary name — for instance by adjoining a ground process $\mathsf{1}$ with $\mathsf{1} \mid \mathsf{1} \mid \mathsf{1} \mid \mathsf{1} \equiv \mathsf{0}$, so that $@\mathsf{1}$ has order four — therefore fails for a reason that is fatal rather than technical: $@R$ and $@(\mathsf{1} \mid R)$ are names of *different addresses*, so no receiver hears both and nothing superposes. Multiplication by i must move a message *between sectors at one address*, not between addresses. This is the requirement that forces the sector to be a coefficient attached to a name rather than a factor inside it, and it is the reason the design expands the namespace instead of enriching the process language.

Remark 1. The four defects are not independent. Repairing the first by group completion supplies the involution needed for the third, and separates the units demanded by the second. Only the fourth is a genuinely separate requirement, and it is the one that dictates *where* the new structure is placed.

3 Syntax

3.1 Sorts

The calculus has three sorts: processes, pairs and names. Pairs are the new sort. A pair is intended as a formal difference of processes, its first component positive and its second negative.

Definition 2 (Grammar)**.**

$$\begin{aligned} P, Q &::= \mathsf{0} \;\mid\; P \mid Q \;\mid\; \mathsf{for}(\langle y_p, y_n\rangle \leftarrow x)P \;\mid\; x!A \;\mid\; *_l x \;\mid\; *_r x \\ A, B &::= \langle P, Q\rangle \\ x, y &::= @_l A \;\mid\; @_r A \end{aligned}$$

Three points of reading. A send carries a pair, not a process. A receipt binds *two* name variables, one per component. And a name is a quotation of a pair, tagged with a sector l or r.

Convention 3 (Drops in process position)**.** $*_s x$ is stipulated above to be a process, whereas the pair quoted by x has two components. We take $*_s x$ to denote the *positive* component of the s-sector content of x, and write $\ominus *_s x$ for the negative component, which is well formed because $\ominus$ is defined on names (Definition 5) and $\ominus *_s x$ abbreviates $*_s(\ominus x)$. No generality is lost and the ordinary reading of $*$ is preserved.

Convention 4 (Abbreviations)**.** We write $x!Q$ for $x!\langle Q, \mathsf{0}\rangle$ and $\mathsf{for}(y \leftarrow x)P$ for $\mathsf{for}(\langle y, y'\rangle \leftarrow x)P$ with y' not free in P. Together with Convention 3 and the choice $@ := @_l$, $* := *_l$, these abbreviations are what make the ordinary rho calculus a sub-language; see Section 7.

3.2 Negation

Definition 5 (Negation)**.** $\ominus$ is defined on pairs and on names by component exchange:

$$\ominus\langle P, Q\rangle := \langle Q, P\rangle, \qquad \ominus @_s A := @_s(\ominus A) \quad (s \in \{l, r\}).$$

It is extended to sends by $\ominus x!A := (\ominus x)!A$.

Proposition 6 (Negation is a total involution on names)**.** *$\ominus$ is defined on every name without reference to reduction, and $\ominus\ominus x = x$.*

Proof. Immediate from component exchange being an involution on ordered pairs. □

Remark 7 (Why the pair is inside the quotation)*.* SETTLED. If pairs occurred only on messages, then names would quote single processes and $\ominus x$ would be undefined: negation would be an operation on messages that the namespace cannot express, so x and $\ominus x$ would not both be addresses, and the polarity clause of the COMM rule (Definition 20) could not be stated. Placing the pair inside the quotation makes negation an operation on addresses and lets the reduction rule mention $\ominus x$ as a name like any other.

There is a second reason, of a different kind. Under the reading in which parallel composition is addition on processes and name multiplication is $@P \cdot @Q = @(P \mid Q)$, quotation is an exponential. Completing to formal differences *before* exponentiating gives a namespace that is the exponential image of a group; completing afterwards does not, and that is why earlier attempts to obtain negation dynamically, as a consequence of annihilation, kept having to produce the sign at reduction time rather than having it available in the syntax.

4 Sectors, polarity, and the group of a name

4.1 Two quotations force two drops

Proposition 8 (Collapse)**.** *Suppose the calculus has two quotation operators $@_l, @_r$ and a* single *dropping operator $*$ satisfying $*(@_s A) = A$ for both s, together with the quote-drop law $@_s(*x) \equiv_N x$. Then $@_l A \equiv_N @_r A$ for every A.*

Proof. Instantiate the quote-drop law at $x := @_r A$ and $s := l$: $@_l A = @_l(*(@_r A)) \equiv_N @_r A$. $\square$

So the two quotations are distinguishable only if the drops are split as well; a map has at most one two-sided inverse. This is why Definition 2 carries $*_l$ and $*_r$ and not a single $*$.

Definition 9 (Sector projection)**.** $*_s(@_t A) := A$ if $s = t$, and $\langle \mathsf{0}, \mathsf{0} \rangle$ if $s \neq t$.

Remark 10. FORK. Definition 9 makes the drops total, so that $*_l$ and $*_r$ behave as real and imaginary part. The alternative — leaving the cross cases stuck — makes the drops partial and puts the sector distinction outside the equational theory, where the matcher cannot use it. Totality is taken here, but the choice interacts with Obligation 5 below and is not innocent.

4.2 The four names of an address

Each name has an underlying pair and two binary attributes: its sector, l or r, and its polarity, which is the choice between a pair and its exchange. Fixing the underlying pair up to exchange therefore leaves four names.

Definition 11 (Address and phase)**.** For $x = @_s A$, write $\mathsf{sec}(x) := s$ and let $\mathsf{addr}(x)$ be the pair A taken up to $\ominus$. The four names sharing an address are

$$@_l A, \qquad @_r A, \qquad @_l(\ominus A), \qquad @_r(\ominus A).$$

Definition 12 (The imaginary unit)**.** SETTLED, with the caveat of Remark 14. Multiplication by i is the successor map on the cycle

$$@_l A \longmapsto @_r A \longmapsto @_l(\ominus A) \longmapsto @_r(\ominus A) \longmapsto @_l A.$$

Equivalently: passing from l to r leaves the pair alone, and passing from r to l exchanges it.

Proposition 13. *The map of Definition 12 has order four, its square is $\ominus$, and the four names of an address form a torsor under $\mathbb{Z}/4$ with $\mathsf{ph}(-)$ the resulting coordinate, $\mathsf{ph}(@_l A) = 1$.*

Proof. Two applications carry $@_l A$ to $@_l(\ominus A)$, which is $\ominus$ by Definition 5; $\ominus$ is an involution by Proposition 6, so the map has order four. The action is free and transitive on the four names by Definition 11. $\square$

Remark 14 (Why i is not the composite of two independent involutions)*.* This is the most delicate point in the design and it must not be glossed. Sector exchange and pair exchange are each involutions, and if they are taken as *independent* the group they generate is the Klein four-group. The Klein four-group is the group of the split-complex numbers, in which $j^2 = +1$ and there are zero divisors, $(1 + j)(1 - j) = 0$. Zero divisors are fatal here, since a pairing would then vanish for reasons having nothing to do with orthogonality. Definition 12 is precisely the stipulation that the two exchanges are *not* independent: the descending passage from r to l carries the pair exchange with it, so that the generated group is cyclic of order four rather than Klein. The asymmetry is not decoration. It is the content of the requirement $i^2 = -1$, and it is what a design with two free involutions cannot supply.

Remark 15 (Only one doubling)*.* Name multiplication is parallel composition, hence commutative. The Cayley–Dickson construction preserves commutativity once and loses it at the second step. A namespace whose product is parallel composition therefore admits exactly one doubling and no more: the calculus can be complex, and cannot be quaternionic, for a structural reason rather than by stipulation.

5 Structural theory

Definition 16 (Structural congruence)**.** $\equiv$ is the least congruence containing alpha-equivalence and satisfying associativity, commutativity and $\mathsf{0}$ as unit for $|$, together with

$$\ominus \ominus A \equiv A.$$

Definition 17 (Name equivalence)**.** $\equiv_N$ is the least congruence satisfying

$$@_s\langle *_s x, \ominus *_s x\rangle \equiv_N x \quad \text{when } \mathsf{sec}(x) = s, \qquad \frac{A \equiv B}{@_s A \equiv_N @_s B}.$$

Remark 18 (The completion equations, and where they live). FORK, and the most consequential one in the note. A pair is intended as a formal difference, which suggests two further equations:

$$\langle Q, Q\rangle \equiv \mathsf{0}, \qquad \langle P, Q\rangle \equiv \langle P \mid R, Q \mid R\rangle.$$

These are what make the pair sort the group completion of processes rather than merely an ordered pair. But $\equiv_N$ contains $\equiv$ under quotation, and COMM fires on $\equiv_N$. Imposing them therefore makes them equations *on addresses*, which the matcher must decide.

The two readings differ sharply.

- **Arithmetic names.** The completion equations hold on names. Cancellation is then real, $\ominus$ is genuine negation, and destructive interference is the matcher discovering that a pair's components have become congruent. The cost is that unique factorisation is lost, the monomial basis with it, and the decidability of name matching becomes a serious question.
- **Coloured names.** The completion equations do not hold on names, which are then a free set with a $\mathbb{Z}/4$ action. Negation is a label carried along and read by the reduction rule, exactly as the sector is. The matcher is untouched; the arithmetic is deferred to a coefficient layer outside the calculus.

The reduction rule of Section 6 never asks whether two differently-signed names are equal, only whether they share an address, so the coloured reading suffices for the operational semantics as it stands. The arithmetic reading is what the semantic programme wants. The note takes neither; it records that the choice determines whether cancellation is a structural event or an external one.

6 Reduction

6.1 Matching is sector-blind and polarity-blind

Definition 19 (Address matching)**.** SETTLED. A receipt at x and a send at x' are *matched* when $\mathsf{addr}(x) = \mathsf{addr}(x')$, irrespective of $\mathsf{sec}(x), \mathsf{sec}(x')$ and irrespective of polarity.

This is forced by the fourth defect of Section 2. If $@_l A$ and $@_r A$ were distinct addresses, then $@_l A!Q \mid @_r A!Q$ would be two messages that no single receiver can hear, rather than a superposition of one message with itself in two sectors; nothing would ever combine, the sector would be a superselection rule, and the doubling would carry no information. Sector and polarity are coefficients on a name, not part of its identity.

6.2 The rule

Definition 20 (COMM)**.** Let $t := \mathsf{sec}(x)$ be the sector of the channel at which the receipt listens, and let $\mathsf{op}(l) := r$, $\mathsf{op}(r) := l$. Then, when the receipt and the send agree in polarity,

$$\mathsf{for}(\langle y_p, y_n\rangle \leftarrow x)P \;\mid\; x!\langle Q_p, Q_n\rangle \;\to\; P\{\, @_{\mathsf{op}(t)}\langle Q_p, \mathsf{0}\rangle/y_p,\;\; @_{\mathsf{op}(t)}\langle Q_n, \mathsf{0}\rangle/y_n \,\}$$

and, when they disagree in polarity,

$$\mathsf{for}(\langle y_p, y_n\rangle \leftarrow x)P \;\mid\; \ominus x!\langle Q_p, Q_n\rangle \;\to\; P\{\, @_{\mathsf{op}(t)}\langle Q_n, \mathsf{0}\rangle/y_p,\;\; @_{\mathsf{op}(t)}\langle Q_p, \mathsf{0}\rangle/y_n \,\}$$

with the symmetric clause for a negative receipt against a positive send. The rule is closed under parallel composition and under $\equiv$ in the usual way.

Two clauses, two effects, and they are independent of one another.

Sector: communication intertwines. SETTLED. The quotation deposited is in the sector *opposite* to the one the receipt listened on. A communication on the real axis deposits an imaginary name; a communication on the imaginary axis deposits a real one. So a communication event is the map $l \mapsto r \mapsto l$, and by Definition 12 it is multiplication by i.

Polarity: opposition exchanges the pair. SETTLED. When the participants have opposite polarity, the two bindings are exchanged, which by Definition 5 is multiplication by $\ominus$. So opposite polarity contributes the sign, and $i^2 = \ominus 1$ is realised as: two sector rotations equal one polarity exchange.

Remark 21 (Phase is path length)*.* An immediate consequence of the sector clause. The sector of a name is the parity of the number of communications that produced it, so the relative phase between two derivations reaching a common outcome is the parity of the difference of their lengths. This is worth naming, because the question of where destructive interference could come from in an unmodified rho calculus had the answer *length-asymmetric reconvergence*, with no mechanism to supply it: concurrency diamonds are symmetric, so the holonomy vanishes. Here the phase *is* the length asymmetry, with no cost functor, no weight map and no external coefficient algebra.

Remark 22 (The identity step is not fictitious)*.* Under the reading of Remark 21, a step that changes no process still rotates the sector. A device introduced elsewhere to complete deficient rows, and justified there case by case, acquires an independent justification here: doing nothing and doing nothing once are genuinely different.

6.3 The four-case refinement

Remark 23. FORK. Definition 20 computes the deposited sector from the receipt's channel alone. But Definition 19 makes matching sector-blind, so a receipt at $@_l A$ may be served by a send at $@_r A$, and the rule as stated ignores the sender's sector. The natural completion is that the two sectors *multiply*:

	send l	send r
recv l	1	i
recv r	i	-1

with -1 realised as $\ominus$ on the deposited pair and the off-diagonal entries being exactly the intertwining of Definition 20. Under this reading the rule of Definition 20 is the off-diagonal half of the table, and $i^2 = -1$ is a case of the reduction rule rather than a separate stipulation.

The refinement is attractive and is not adopted here, because it changes what the rule computes rather than extending it: on the diagonal it deposits a name in the sector it received on, which contradicts the plain reading of "communication intertwines". Settling this is the first item of Section 9.

Remark 24 (Amplitude is multiplicity)*.* Under Definition 19, the expression $@_l A!Q \mid @_r A!Q$ is a single address carrying one copy in each sector, and with the pair supplying signs this is $(1+i)$ times a send at A. Generally, the coefficient of a message is the multiset of sectors and polarities its copies occupy, so Gaussian-integer coefficients are realised as multiplicity under parallel composition, with no external coefficient algebra and no tagging. This is the payoff that motivated Definition 19, and it should be the first thing checked against any proposed revision of the rule.

7 Conservativity

Proposition 25 (Embedding)**.** *FORK, stated as a target rather than a theorem. Let $\iota(-)$ send an ordinary* rho *term to the term of Definition 2 obtained by*

$$\iota(x!Q) := \iota(x)!\langle \iota(Q), \mathsf{0}\rangle, \qquad \iota(@P) := @_l\langle \iota(P), \mathsf{0}\rangle,$$
$$\iota(\mathsf{for}(y \leftarrow x)P) := \mathsf{for}(\langle y, y'\rangle \leftarrow \iota(x))\iota(P), \qquad \iota(*x) := *_l\iota(x)$$

with y' fresh, and homomorphically elsewhere. Then $\iota(-)$ is injective and preserves $\equiv$.

Remark 26 (Conservativity fails for reduction, and deliberately)*.* The image of $\iota(-)$ is not closed under $\to$. A single communication moves a name off the real axis, so an ordinary rho program is not, in this calculus, a program that stays classical. This is not an oversight to be repaired: it is the content of Remark 21, since a calculus in which ordinary programs remained on the real axis would be one in which no phase ever accumulated. The design is a genuinely different calculus, not an extension, and it should be presented as such.

If a conservative reduction is wanted, the diagonal entries of the table in Section 6.3 supply it: under that reading, a real receipt served by a real send stays real, and the ordinary calculus embeds as the real sub-calculus. That is an argument for the refinement, and is recorded here rather than in Section 6.3 because it is the strongest one.

8 The dagger, in one paragraph

The construction this calculus serves requires an antilinear involution on states. Two facts from the design discussions fix its shape and are recorded here so that the calculus can be checked against them, though the dagger itself is not developed in this note.

First, the dagger is not a syntactic polarity swap on term formers. It is obtained by residuation against a pole: $Q^\dagger$ is a witness of $\psi_Q \rhd \bot$, where $\rhd$ is the adjoint to the separating conjunction in the generated spatial logic and $\bot$ is a behavioural formula, annihilation being the candidate. This makes the dagger a property first and a term second, hence stable under any presentation generating the same logic.

Second, the sector and polarity components of the dagger are now fixed. Flipping both sectors is the map $(a, b) \mapsto (b, a)$, which in the complex numbers is $z \mapsto i \cdot \bar{z}$ and not conjugation; taken as the dagger it makes the pairing skew-Hermitian and puts $\langle P \mid P \rangle$ in the imaginary sector, so positivity fails outright. The correction is to compose the sector flip with a pair exchange,

$$@_l \mapsto \ominus @_r, \qquad @_r \mapsto \ominus @_l,$$

which is $z \mapsto -i\bar{z}$, is involutive, and is assembled entirely from operations Definitions 5 and 12 already provide.

Remark 27 (The positivity obligation, made precise). Since every communication rotates the sector and the dagger rotates it once more, the sector in which $\langle P \mid P \rangle$ lands depends on the number of communications consumed in the annihilation of $P^\dagger$ against P, modulo four. Positivity therefore requires that all annihilation paths of $P^\dagger \mid P$ agree modulo four. This is a graded confluence condition, weaker than confluence, and it is not obviously true. It is cheap to test on the ladder of mutually annihilating names, and it should be tested before anything further is built on the dagger.

9 Obligations

Obligation 1 (The four-case table)**.** Settle whether the deposited sector is computed from the receipt's channel alone (Definition 20) or from both participants (Section 6.3). Remark 26 is the strongest argument for the latter. This decision governs everything downstream and should be taken first.

Obligation 2 (Arithmetic or coloured names)**.** Settle Remark 18: whether the completion equations hold on names, and hence whether cancellation is a structural event decided by the matcher or an external one deferred to a coefficient layer.

Obligation 3 (Decidability of name matching)**.** COMM fires on $\equiv_N$. Under the arithmetic reading of Remark 18, the matcher must normalise formal differences nested through quotation. Termination is not obvious, and this is the practical gate on the entire design.

Obligation 4 (Freshness)**.** The property that a quotation of P is not free in P is what obviates a freshness operator and underwrites freshness by quotation. Its proof proceeds by structural complexity. Both quotation operators must increase that measure, which is routine; but the quote-drop law of Definition 17 relates a name to a term built from its own drops, and the completion equations, if adopted, shorten terms non-structurally. The mutual recursion of $\equiv$ and $\equiv_N$ was already the delicate part of the metatheory. That argument must be redone rather than adapted.

Obligation 5 (Observability of the sector)**.** $*_l$ and $*_r$ are term formers, so any context may apply both to a name and see which yields $\mathsf{0}$. The sector is therefore classically readable, which makes it a superselection rule rather than a phase, and makes global phase detectable by a two-line context. The natural remedy is to admit the drops in guard position only, where evaluation is pure and non-consuming, and to let term-position dropping be $*_l$ by convention. This needs deciding, and it interacts with Definition 9.

Obligation 6 (Scalar extraction and occurrence counts)**.** For a race to interfere, its candidates must reach the same outcome with different coefficients. Consuming from $@_l A$ rather than $@_r A$

yields $P\{@_r Q/y\}$ rather than $P\{@_l Q/y\}$, and these are the same outcome up to phase only if the sector can be extracted from the name to the term, that is, only if there is a homomorphism from processes to $\mathbb{Z}/4$ under which substitution multiplies. The natural candidate is sector degree, the product of sectors over name occurrences — but then the phase is raised to the power of the number of occurrences of y, so extraction holds only when y occurs exactly once. Occurrence count is the exponent on the phase. Either linear use of names bound from graded channels is required, or multiplicative phase in occurrences is accepted and scalar extraction abandoned. This is a linearity requirement of a different kind from the ones previously considered and declined, and the earlier reasons for declining do not apply to it.

Obligation 7 (Positivity)**.** Discharge or refute Remark 27: that all annihilation paths of $P^\dagger \mid P$ agree in length modulo four.

Obligation 8 (Two gradings on names)**.** This calculus grades names by sector. A separate line of work grades them by binding structure, unforgeable names being those introduced by a name-restriction operator and classical ones being quotations. The two gradings are independent, so there are four kinds of name, and their interaction is unstated. The conjecture worth testing is that sectors should live only in the unforgeable namespace, which would also confine Obligation 5.

Obligation 9 (Metatheory)**.** Nothing in this note is proved beyond Propositions 6, 8 and 13. Subject reduction for the sector grading, whether bisimulation is a congruence in the presence of two quotations, and whether the sector is visible to bisimulation at all, are all open. The last is pressing: an inert distinction that $\equiv$ can see and $\sim$ cannot is exactly the configuration in which the spatial layer strictly refines behaviour, and the sign structure would then be behaviourally invisible.

10 What this note does not do

It does not give the interpretation map, the pairing, or the evaluation operation against which the pairing is defined. It does not establish that the resulting scalars form a rig, only that the four obstructions to their doing so have each been given a place to be repaired. It does not address summability when the set of terminal states is infinite. It does not treat the join, whose generalisation to graded messages requires a sum over matchings with coefficients in a group ring, and whose normalisation may or may not remain the right unit. And it takes no position on whether the resulting calculus is worth the cost of Obligation 3, which is the question a reader should keep in view throughout.

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